

Zhimin Li

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Education

University of Pittsburgh, Pittsburgh, PA, USA	Jan. 2024–Present
Ph.D. in Electrical and Computer Engineering	
Northeastern University, Boston, MA, USA	Jan. 2023–Jan. 2024
M.S. in Information Science (transfer to University of Pittsburgh)	
University of Pittsburgh, Pittsburgh, PA, USA	Sept. 2019–Dec. 2022
B.S. in Computer Science	
Hefei University of Technology, Hefei, Anhui, China	Sept. 2017–Jun. 2019
B.S. in Architectural Environment and Energy Application Engineering (transfer to University of Pittsburgh)	

Research Interests

Efficient and Trustworthy AI; Edge/Embedded AI; Vision-Language Models; DNN Acceleration; AI Fairness and Privacy.

Research Experience

Privacy-Preserving Wearable Vision-Language Models Nov. 2025–Present

Advisor: Jingtong Hu, Pittsburgh, PA, USA

- Investigated a practical privacy risk in wearable AI (smart glasses): even when a VLM produces a harmless answer, its intermediate visual representations can still expose sensitive attributes such as location, income, sex, and personal interests.
- Designed a split-inference privacy architecture that transforms visual tokens **before they leave the trusted wearable device**, requiring joint optimization of representation disentanglement, semantic privacy suppression, and utility preservation across multiple VLM backbones.
- Built a **3,221-record privacy–utility benchmark** and reduced private-attribute inference accuracy from **56.7% to 7.4%** while retaining **74.4% utility**. Manuscript currently under review at **NeurIPS 2026**; see [related publication below](#).

Achieve Fairness in AI-Assisted Mobile Healthcare Apps Nov. 2023–Nov. 2025

Advisor: Jingtong Hu, Pittsburgh, PA, USA

- Addressed demographic bias in medical AI, where models trained on imbalanced clinical datasets can perform well on average while systematically underperforming for underrepresented patient groups.
- Developed a fairness-oriented medical imaging pipeline spanning data generation, preference alignment, model optimization, and downstream diagnosis; incorporated clinician-informed preferences so that generated data remained both demographically balanced and clinically meaningful.
- Co-developed **FairGen**, improving fairness by **95.9%** on skin images, **80.0%** on chest radiography, and **35.2%** on brain MRI while maintaining competitive diagnostic accuracy. Published in **npj Digital Medicine** (IF 18); see [related publication below](#).

Efficient Framework for Real-Time Object Detection on the Edge Mar. 2023–Nov. 2023

Advisor: Yanzhi Wang, Boston, MA, USA

- Tackled the gap between high-accuracy object detection models and the strict power and latency budgets of mobile devices, where processing every video frame at maximum hardware frequency wastes substantial computation and energy.
- Contributed to an adaptive deployment framework that jointly coordinates **video keyframe selection, CPU/GPU DVFS, and DNN pruning** instead of optimizing these components independently, creating a large hardware–model design space under real-time latency constraints.
- Maintained object-detection accuracy within **0.28% mAP** of processing every frame while reducing system power consumption by up to **96.7%**. Accepted by **ICCAD 2024**; see [related publication below](#).

Digital Avatar on Mobile Device Aug. 2023–Nov. 2023

Advisor: Yanzhi Wang, Boston, MA, USA

- Explored how multiple generative-AI components can be integrated into a responsive digital human rather than used as isolated models, with the goal of enabling natural real-time human–AI interaction on mobile clients.
- Built an end-to-end server-to-mobile pipeline connecting **speech recognition, LLM reasoning, text-to-speech, facial animation, and audio–video streaming** through FastAPI, requiring synchronization across heterogeneous models and different inference latencies.
- Evaluated generated personalities through human preference studies; the proposed prompting strategy received **58.3% of votes** as the most humorous response. Accepted to the **IJCAI 2024 Demo Track**; see [related publication below](#).

DNN Data Management on Hybrid Memory Systems Feb. 2021–Sep. 2021

Advisor: Dazhao Cheng, Charlotte, NC, USA

- Studied a systems bottleneck that becomes increasingly important as DNN models grow: fast memory offers high performance but limited capacity, while larger memory tiers introduce substantially higher data-access cost.

- Developed and evaluated data-management strategies for placing and migrating DNN data across heterogeneous memory tiers, requiring coordination between model execution, memory behavior, and runtime data movement rather than treating memory as a uniform resource.
- Contributed to system evaluation and performance analysis using ReDL; the work was accepted by **IEEE Big Data 2021**. See [related publication below](#).

Publications

1. **Zhimin Li***, Ruichen Zhang*, Zhen Tan*, Howard J. Aizenstein, Jingtong Hu, and Tianlong Chen. “FairGen: Preference-Aligned Diffusion for Demographically Equitable Medical Image Synthesis.” *npj Digital Medicine*, **IF 18.0**, 2026. [doi:10.1038/s41746-026-02868-z](#). *Equal contribution.
2. Chao Wu, Yifan Gong, Liangkai Liu, Mengquan Li, Yushu Wu, Xuan Shen, **Zhimin Li**, Geng Yuan, Weisong Shi, and Yanzhi Wang. “AyE-Edge: Automated Deployment Space Search Empowering Accuracy yet Efficient Real-Time Object Detection on the Edge.” In *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2024, Art. 178, pp. 1–9. [doi:10.1145/3676536.3676655](#).
3. Timothy Rupprecht, Sung-En Chang, Yushu Wu, Lei Lu, Enfu Nan, Chih-hsiang Li, Caiyue Lai, **Zhimin Li**, Zhijun Hu, Yumei He, David Kaeli, and Yanzhi Wang. “Digital Avatars: Framework Development and Their Evaluation.” In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*, Demo Track, 2024, pp. 8780–8783. [doi:10.24963/ijcai.2024/1031](#).
4. Wei Rang, Donglin Yang, **Zhimin Li**, and Dazhao Cheng. “Scalable Data Management on Hybrid Memory System for Deep Neural Network Applications.” In *2021 IEEE International Conference on Big Data (Big Data)*, 2021, pp. 1470–1480. [doi:10.1109/BigData52589.2021.9671309](#).

Under Review

5. **Zhimin Li**, Pan Wang, Jingxian Chen, Yuantao Tang, Anthony Chen, Qian Lou, and Jingtong Hu. “Defending Wearable VLMs Against Private Attribute Inference.” [arXiv:2608.28691](#), 2026. Under review at NeurIPS 2026.
6. Jingxian Chen, Shengjie Xu, **Zhimin Li**, Pan Wang, Samarjit Chakraborty, Jingtong Hu. “Safety-Aware Differentiable DNN Search for CPS Perception.” Manuscript under review.
7. Pan Wang, Jie Peng, **Zhimin Li**, Jingxian Chen, Hui Ji, Huanrui Yang, Pengfei Zhou, Tianlong Chen, Jingtong Hu. “Think Before You Compute: Adaptive Budgeting and Compression for Efficient MoE-Based Multimodal Learning.” Manuscript under review.

Work Experience

Northeastern University, Research Assistant, Electrical and Computer Engineering *Sep. 2023–Dec. 2023*

- Worked on DNN weight pruning, DVFS for mobile DNNs, and deployment of TTS, Stable Diffusion v1.5/v2.0, and object detection models on mobile devices.

University of Pittsburgh, Research Assistant, Electrical and Computer Engineering *May 2022–Dec. 2023*

- Parallelized CPU–FPGA DRAM data transfer with CNN inference/training and optimized the accelerator for higher throughput and lower latency.

University of Pittsburgh, Teaching Assistant, SCI–Computer Science *Fall 2022–Present*
CS 445: Data Structures; ECE 1110: Computer ORG and Architecture; ECE 1150 Computer Network.

Honors & Academic Service

Reviewer: CVPR, IJCAI, ICCAD, DAC, IEEE Transactions on Cyber-Physical Systems (TCPS).

Dean’s List: 2022 Summer, 2022 Spring, 2021 Summer, 2020 Fall, 2020–2021 Spring.

Skills

Programming & Systems: Python, Linux, CUDA, Docker, Git, FastAPI.

Machine Learning & Applied AI: PyTorch, Vision-Language Models (VLMs), Multimodal Learning, Generative AI, Diffusion Models, Object Detection, Model Compression, Privacy-Preserving ML, AI Fairness.

AI Infrastructure & Edge AI: ML Systems, Model Inference, Inference Orchestration, On-Device Inference, Split Inference, GPU Computing, DNN Acceleration, DVFS, FPGA Acceleration, Hybrid Memory Systems.

Models & Frameworks: Qwen, MiniCPM, GLM, Stable Diffusion, YOLOv5, nanoGPT, Whisper.